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Endotherms trade body temperature regulation for the stress response

Joshua K. R. Tabh[1,3], Mariah Hartjes[2] and Gary Burness[2]

[1]Environmental and Life Sciences Graduate Program, and [2]Department of Biology, Trent University, Peterborough, Ontario, Canada K9L 0G2
[3]Department of Biology, Lund University, Lund, 223 62, Sweden

JKRT, 0000-0002-9519-7488; GB, 0000-0002-1695-7179

Research

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Author for correspondence:
Joshua K. R. Tabh
e-mail: joshua.k.tabh@gmail.com

Responding to perceived threats is energetically expensive and can require animals to curtail somatic repair, immunity, and even reproduction to balance energy ledgers. In birds and mammals, energetic demands of thermoregulation are often immense, yet whether homeostatic body temperatures are also compromised to aid the stress response is not known. Using data sourced from over 60 years of literature and 24 endotherm species, we show that exposure to non-thermal challenges (e.g. human interaction, social threats) caused body temperatures to decrease in the cold and increase in the warmth, but particularly when species-specific costs of thermoregulation were high and surplus energy low. Biophysical models revealed that allowing body temperature to change in this way liberated up to 24% (mean = 5%) of resting energy expenditure for use towards coping. While useful to avoid energetic overload, these responses nevertheless heighten risks of cold- or heat-induced damage, particularly when coincident with cold- or heatwaves.

1. Introduction

About 2000 years ago, the Greek physician Galen observed that the body temperatures of his patients tended to rise after psychologically distressing experiences [1]. Given the acute nature of these rises, Galen termed them 'ephemeral fevers' [2] and speculated, like those after him (e.g. Ibn Sina), that they emerged from an abundance of humour [3,4]. 'Why' (i.e. functionally) they might emerge at all, however, was apparently left unquestioned.

Numerous studies have now shown that Galen's 'ephemeral fever' is not just restricted to humans. Instead, many rodent, non-hominid primate, and even bird species appear to change their body temperatures after exposure to common disturbances, such as transportation or restraint [5–7]. Moreover, accumulating evidence suggests that the term 'ephemeral fever' is likely a misnomer, with body temperatures of some individuals *declining* in response to some stressors rather than rising [8,9]. That these stress-induced thermal responses exist across varied taxonomic groups and may differ under certain conditions strongly intimates a possible function, or, perhaps, a conserved and inherent nature (i.e. a constraint) behind them.

Recently, we proposed that changes in body temperature following stress exposure may be understood as consequences of an allocation trade-off between thermoregulation and the stress response ([10,11]; similar to [12]). Under this hypothesis, fluctuations in body temperature that accompany the stress response represent the outcomes of reducing energy expenditure towards thermoregulation in favour of expenditure towards coping. Unlike previous hypotheses, this particular hypothesis provides an explanation for why body temperature might either rise *or* decline after a stress exposure, since relaxing thermoregulation should merely result in body temperature drifting towards ambient (particularly in the cold) or in the direction of accumulating metabolic

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